INTERPRETABILITY-GUIDED CIRCUIT ROUTING FOR HINDI-ENGLISH LORA FINE-TUNING

Team BabyShark

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motivation

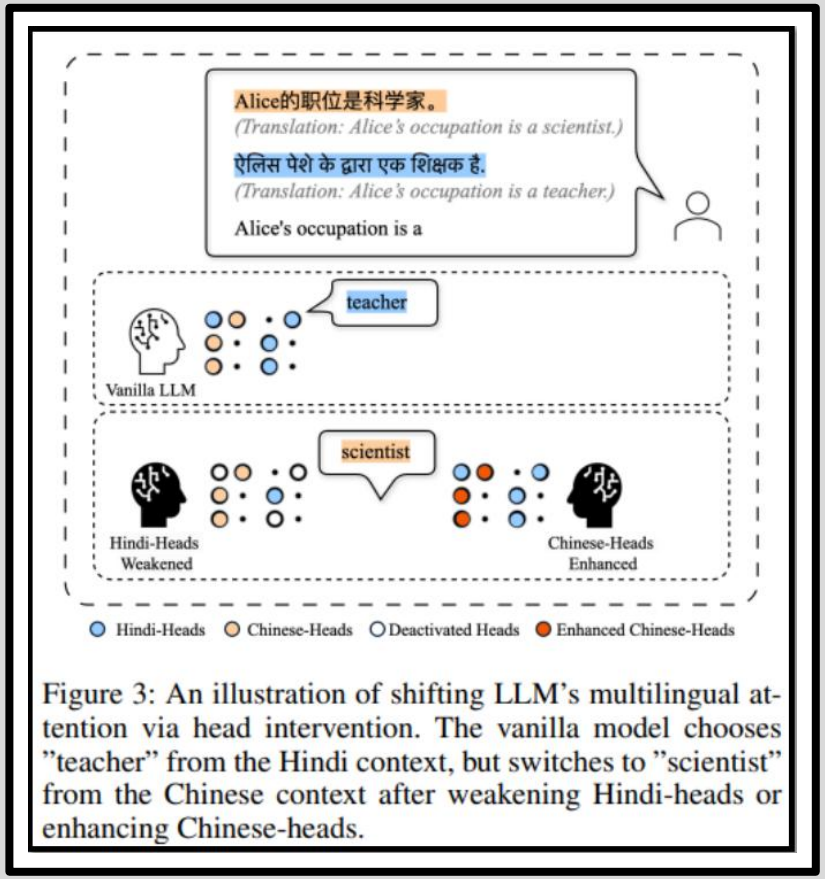


Figure 3: An illustration of shifting LLM's multilingual attention via head intervention. The vanilla model chooses "teacher" from the Hindi context, but switches to "scientist" from the Chinese context after weakening Hindi-heads or enhancing Chinese-heads.

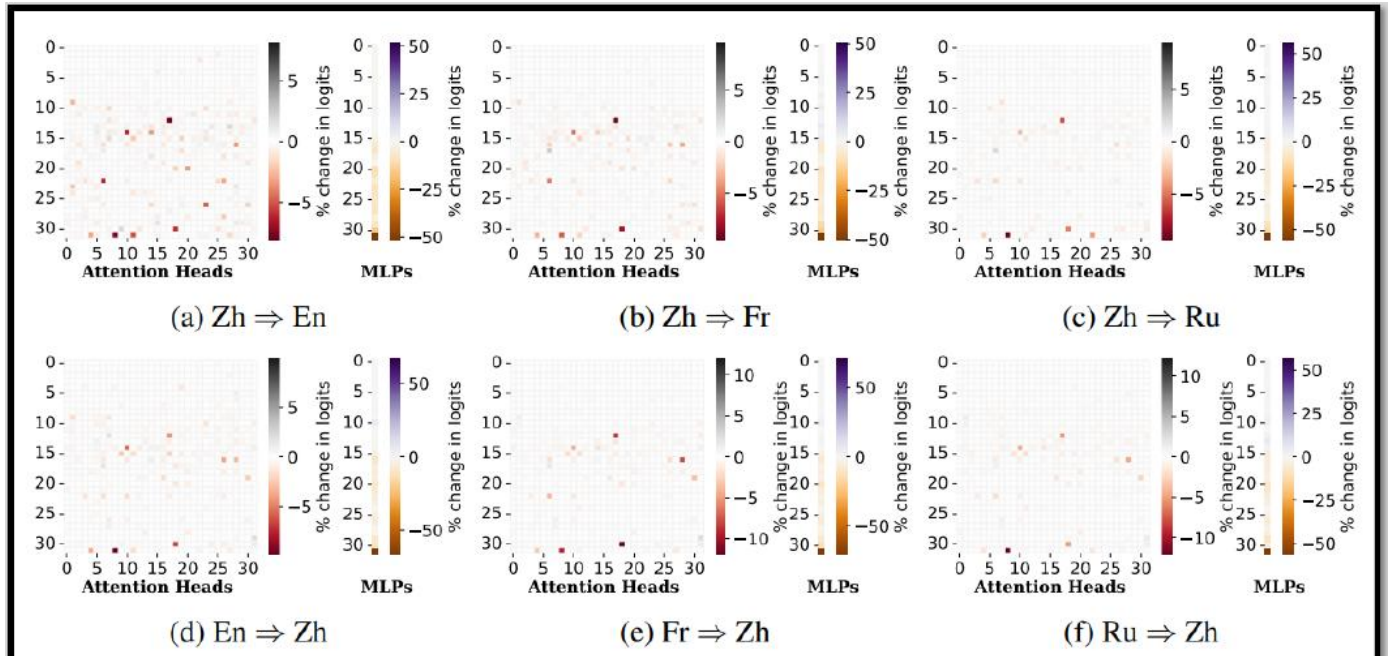


Figure 1: Importance of heads related to translation across different directions. Each square at position (x, y) refers to the x-th head in the y-th layer. Red (Brown) squares denote heads (MLPs) that have a positive impact on predicting the target token, while grey (purple) squares indicate heads (MLPs) with a negative effect. Additional explanations of this figure are available in Apdx. D.3.

Liu et al., Focusing on Language (AAAI 2026): weakening Hindi heads or enhancing Chinese heads shifts the model's language.

Zhang et al., Exploring the Translation Mechanism of LLMs (NeurIPS 2025): translation relies on few heads and specific MLPs. Figures from the cited papers.

Our question: can causal patching scores tell us where to place a LoRA budget?

research question and hypotheses

Setup: Qwen2.5-1.5B, 28 layers × 12 query heads = 336 heads, plus 28 MLP layers. Task: Hinglish Hinglish sentiment (SAIL-2017).

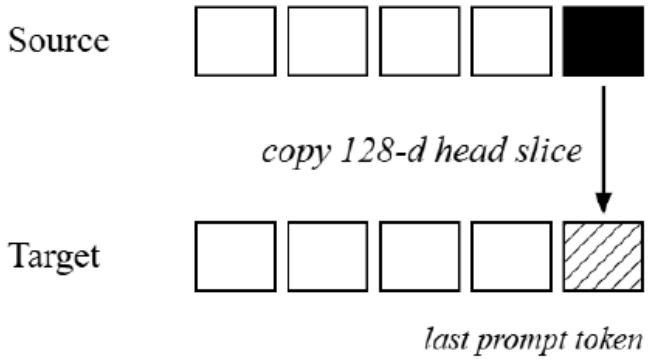
H1 localisation: a few heads and MLP layers carry the language and script contrasts.

H2 independence: heads act additively, so single-head scores can rank sets of heads.

H3 routing: with equal active units, routing by our scores beats random routing (heads, MLP layers, both).

H4 efficiency: a routed adapter comes close to uniform LoRA with fewer mask-active parameters.

methodology: activation patching



(a) Patching a single head

Contrast	Source → target	Held fixed
Language-only	`hi_latn → en`	script
Script-only	`hi_deva → hi_latn`	language
Both	`hi_deva → en`	none

(b) The three contrasts

$$R_{l,h} = \frac{m_{\text{patch}} - m_{\text{corr}}}{m_{\text{clean}} - m_{\text{corr}}}$$

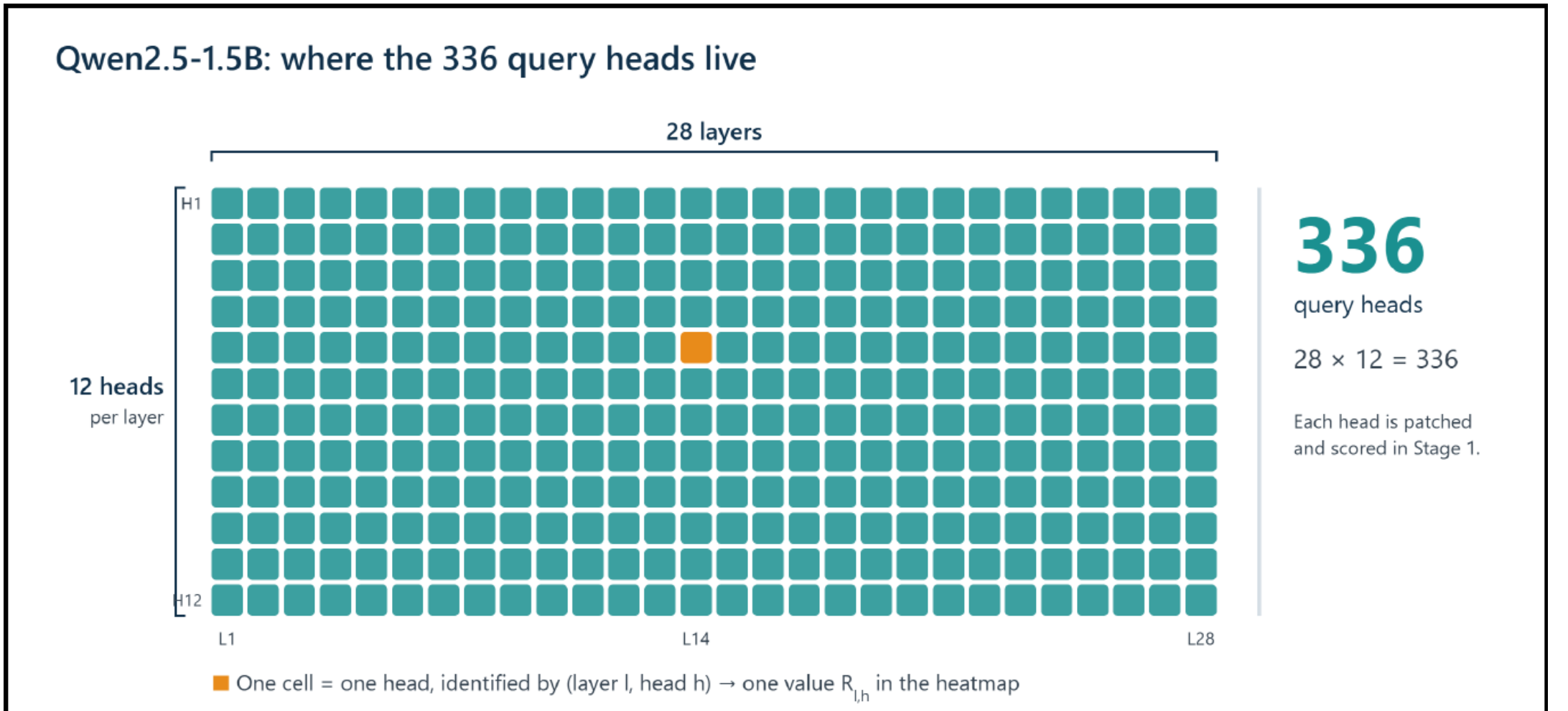
$$38 \times 336 \times 3 \times 2 = 76{,}608$$

forward passes

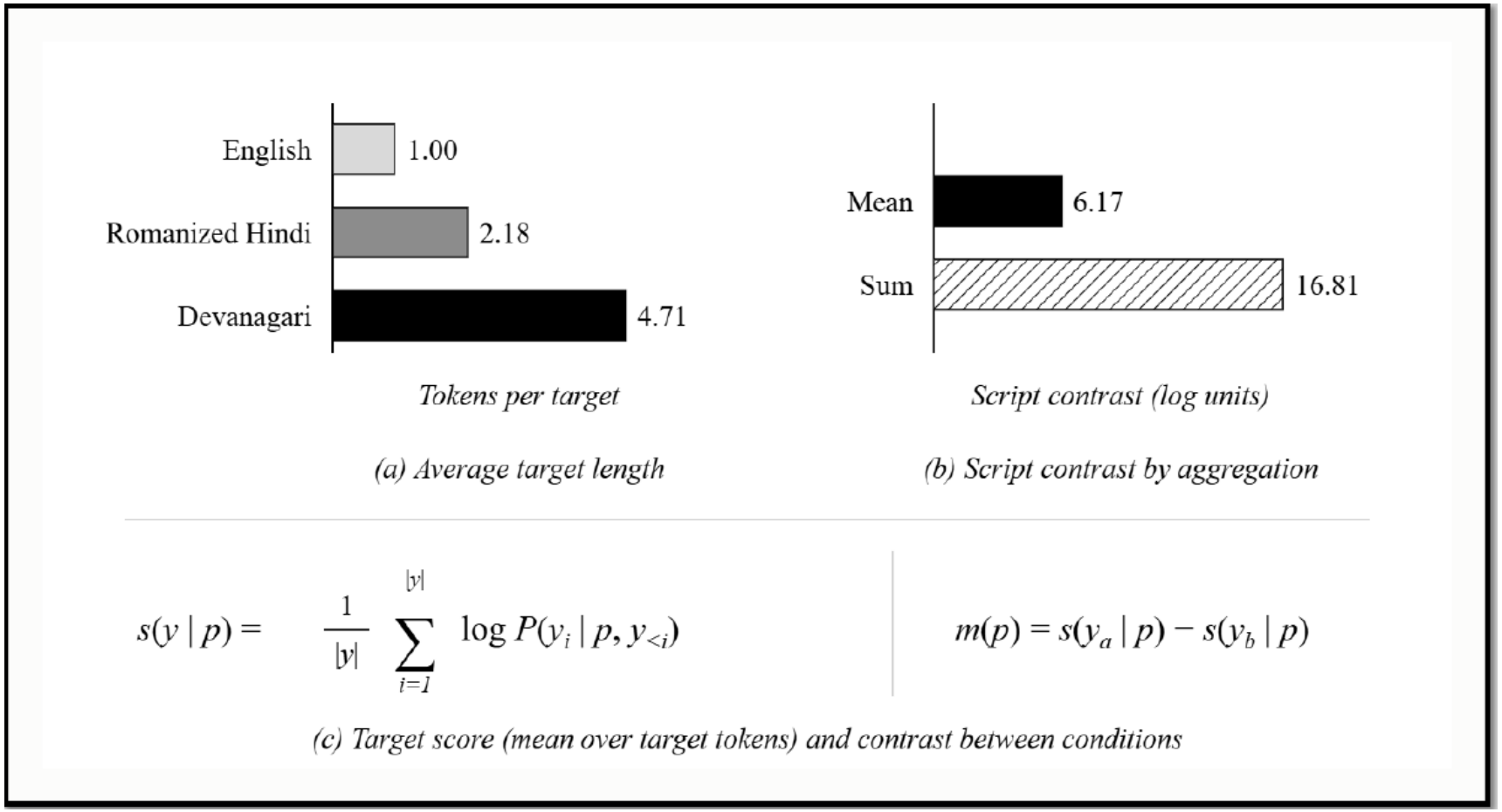
(c) Recovery (1 = full restoration, 0 = none) and total number of forward passes

Figure from our mid-submission deck: pilot set of 38 concepts. The primary set has 39, i.e. 39 × 336 × 3 × 2 = 78,624 head passes.

methodology: the 336 query heads

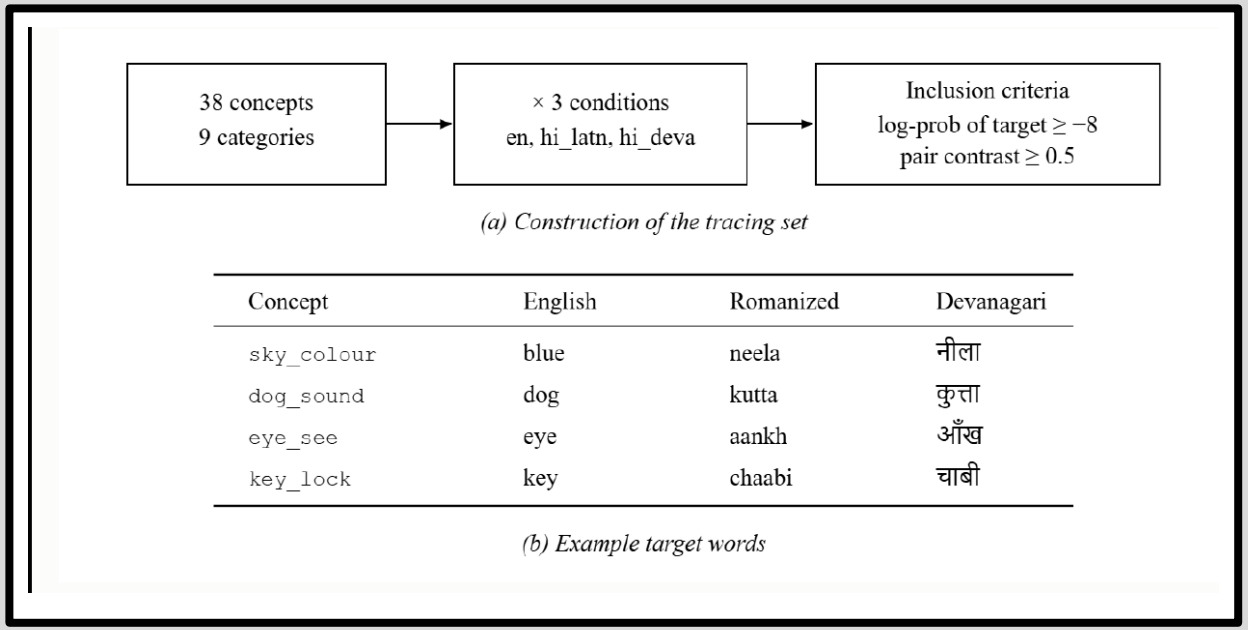


methodology: scoring and contrast



Devanagari targets take 4.71 tokens vs 1.00 for English, so we score by mean per-token log-probability, not the sum.

methodology: concept sets



(a) Construction of the tracing set

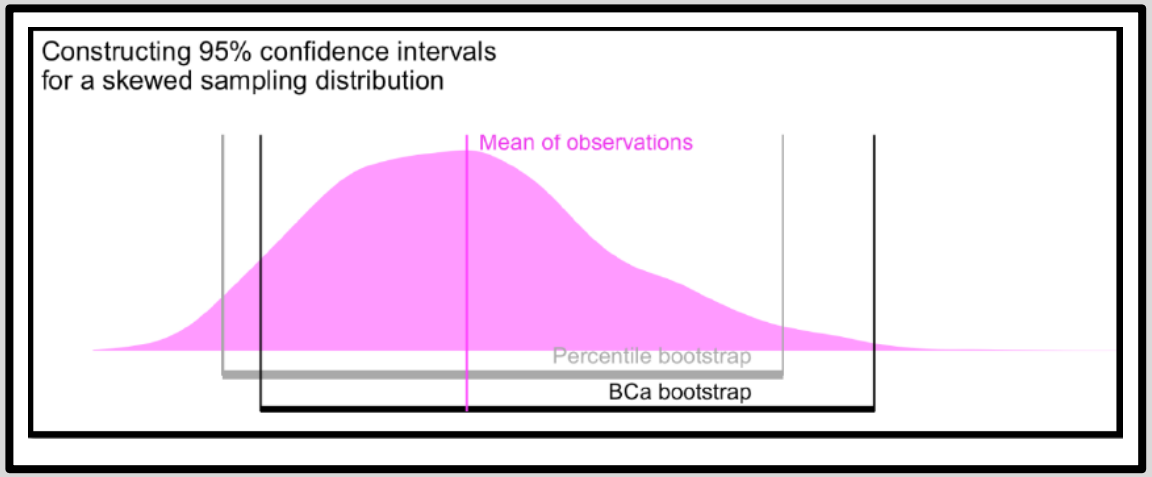
Concept	English	Romanized	Devanagari
sky_colour	blue	neela	नीला
dog_sound	dog	kutta	कुत्ता
eye_see	eye	aankh	आँख
key_lock	key	chaabi	चाबी

(b) Example target words

- Hindi WordNet + IndoUKC, 20 semantic/part-of-speech groups
- Each concept: English, Romanized and Devanagari prompt with the target word
- Sets overlap, so they are sensitivity checks

Set	**Concepts**	**Role**
Linguistic (primary)	39 of 50 (thresholds −8, 0.5)	Main results
Loosened thresholds	45 of 50	Sensitivity
Pilot (figure above)	38 concrete, 9 categories	Sensitivity
Larger set	120 (99 pass thresholds)	Sensitivity, overlaps main

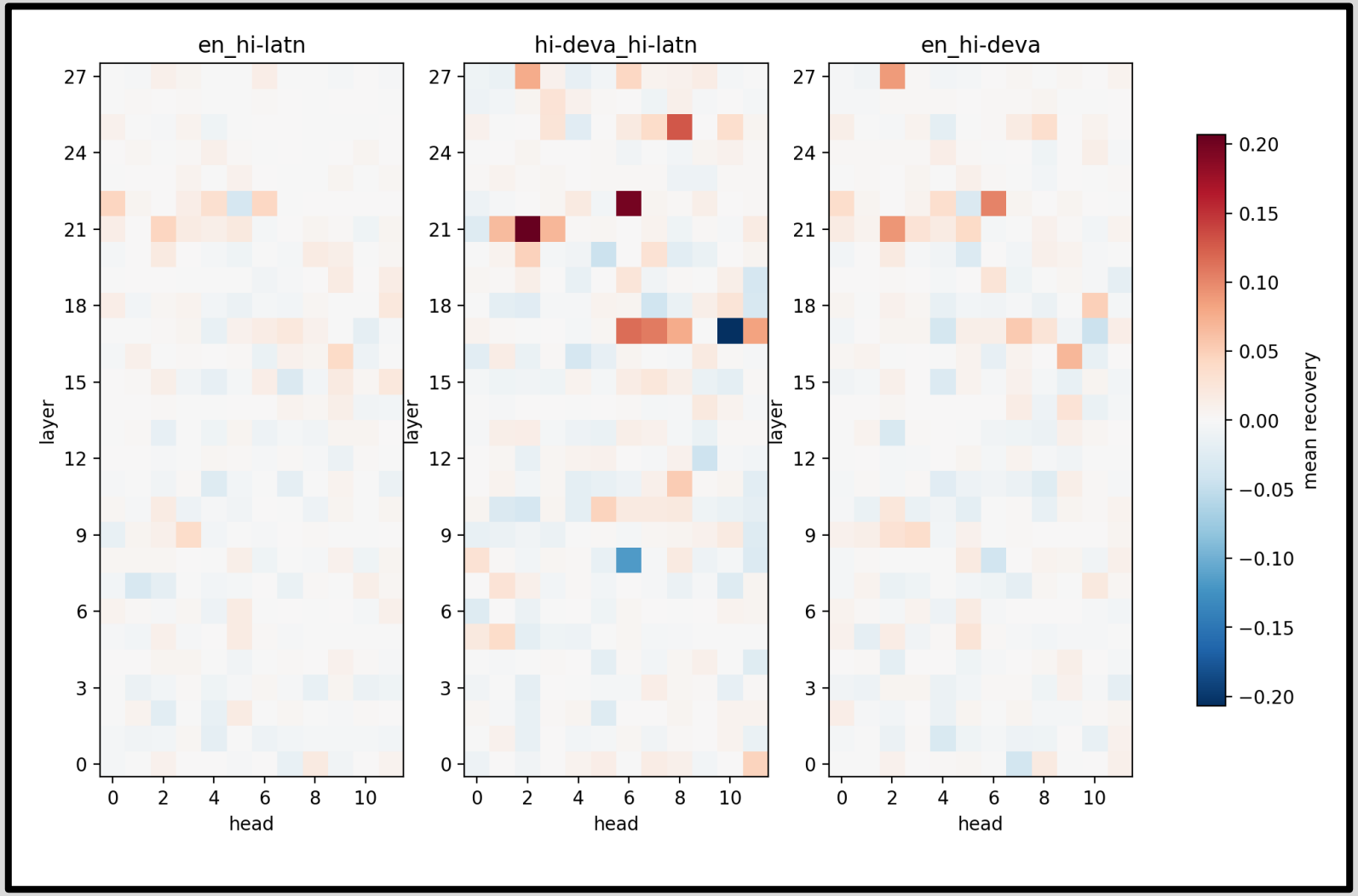
stage 1: significance testing



- Percentile bootstrap CIs over concepts (2,000 resamples)
- Sign-flip max-statistic test for family-wise error over 336 heads, 28 layers, 1,225 pairs
- ~8.4 intervals exclude zero by chance; 116 do, so CIs alone are not enough
- Leave-one-concept-out check per top head

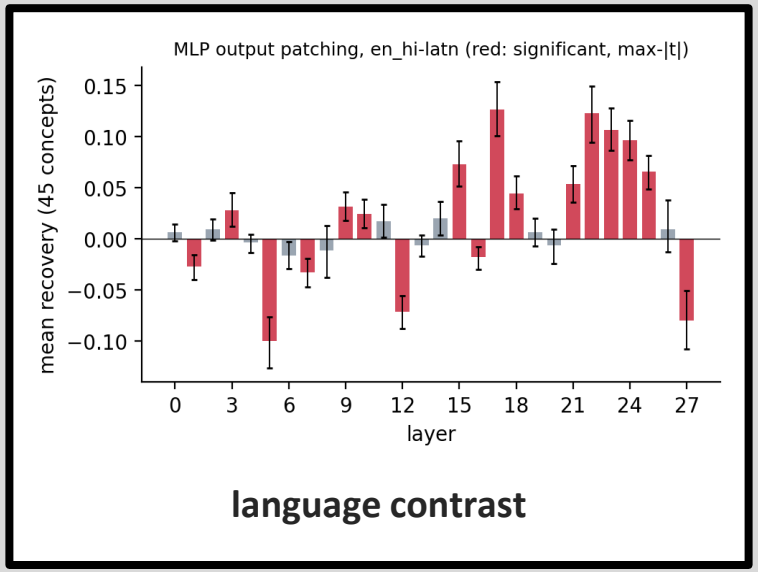
Language contrast	Result
Heads passing FWER	14 of 336
Top heads	L22H0 0.044, L21H2 0.043, L22H6 0.039
MLP layers significant	15 of 28
Top MLP layers	L17 0.131, L22 0.117, L23 0.110

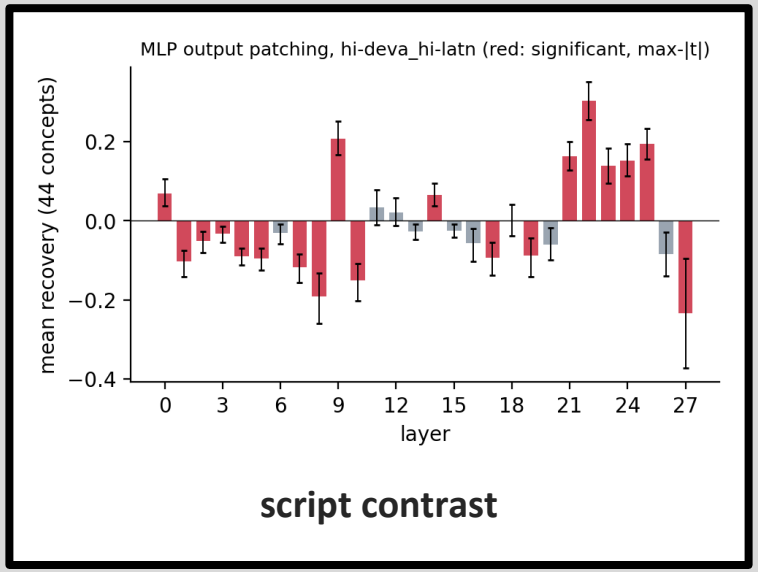
stage 1: where language and script live

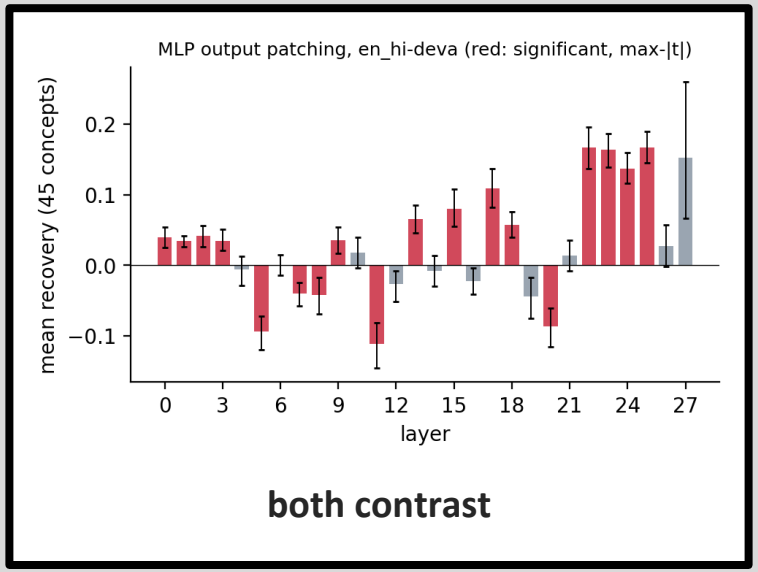


- A small group of late heads: L21H2, L22H6, L22H0
- Late MLP layers recover more than any single head
- H1 supported

stage 1: mlp layers carry more

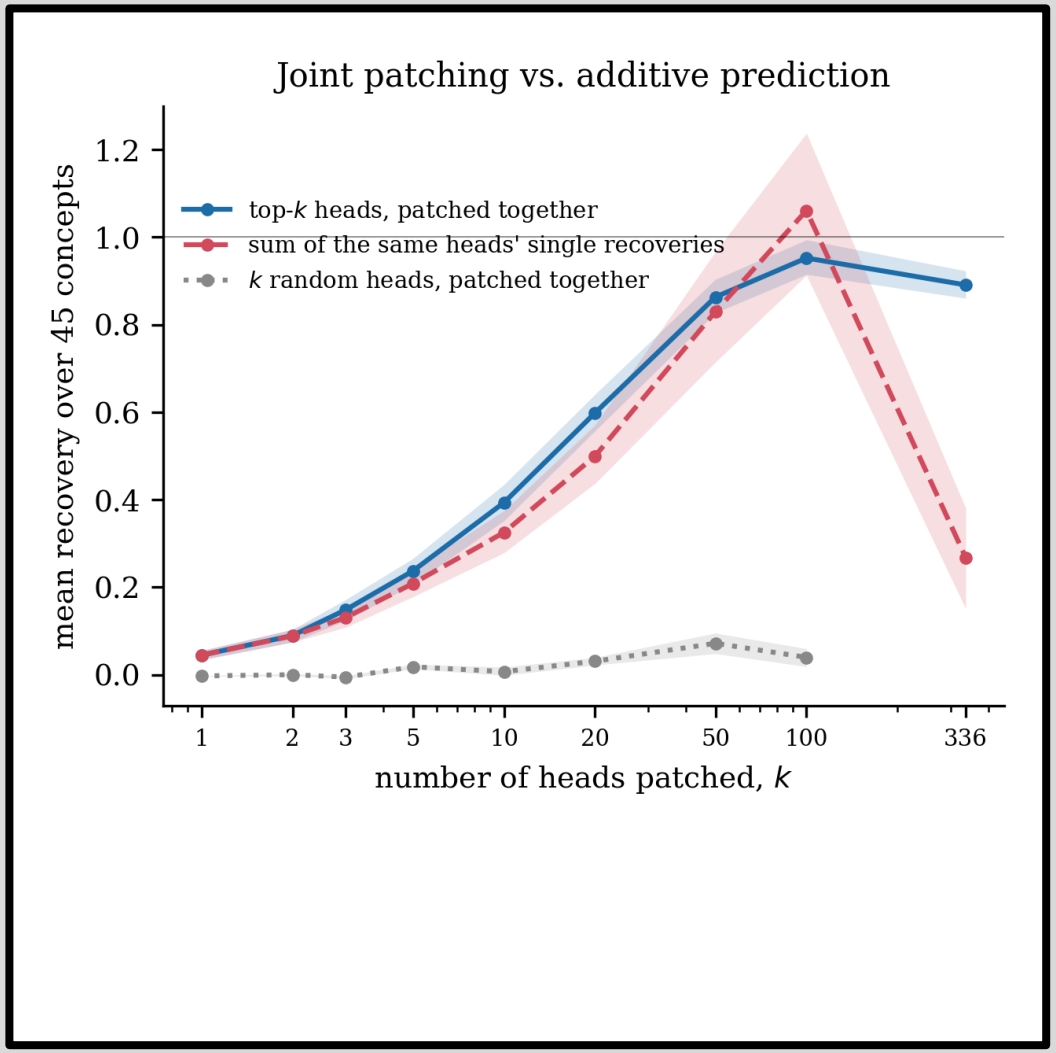






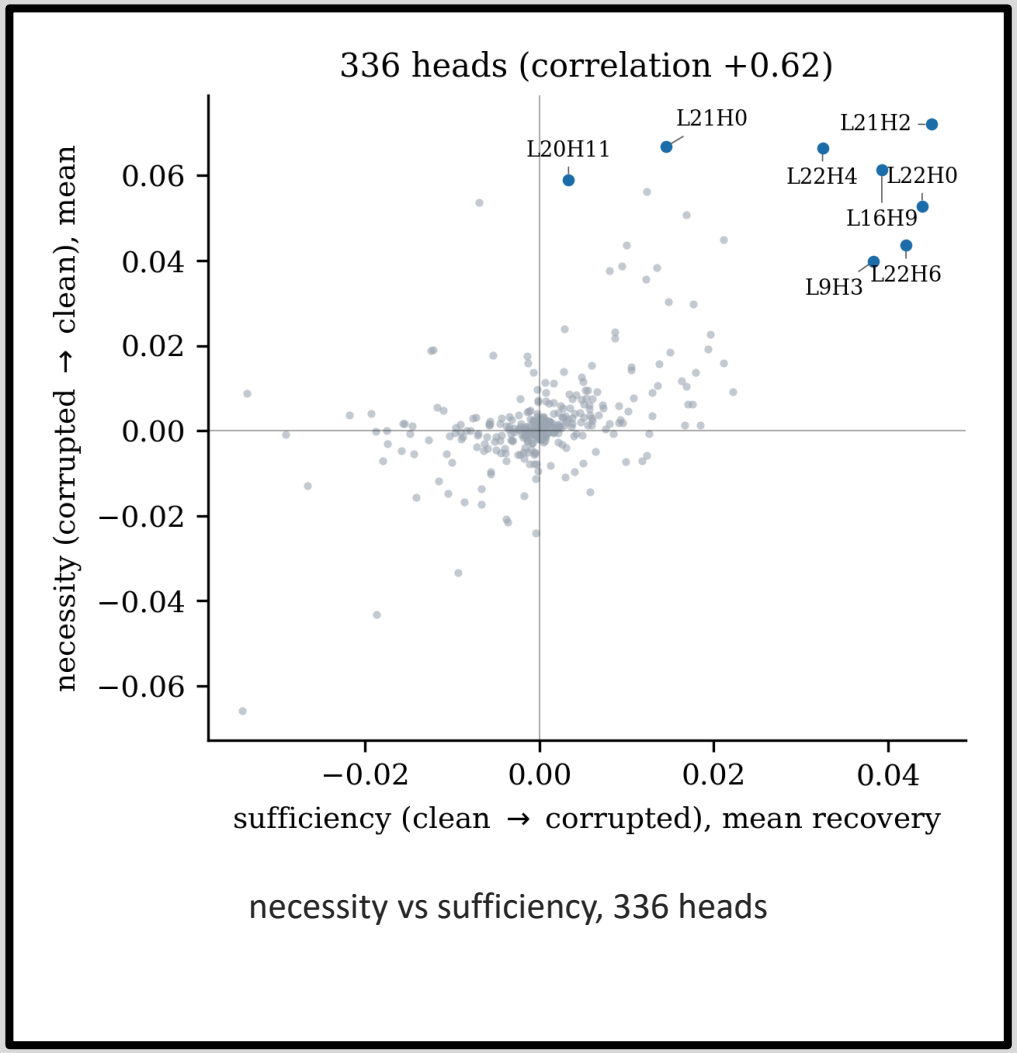
- Language: 15 of 28 layers significant; top L17 0.131, L22 0.117, L23 0.110
- Script: 18 significant; top L22 0.318, L9 0.217, L25 0.193
- Both: 17 significant; top L27 0.175, L25 0.172, L23 0.171
- Same late layers as the leading heads; a whole MLP block is a much larger unit than one head

stage 1: heads are not independent

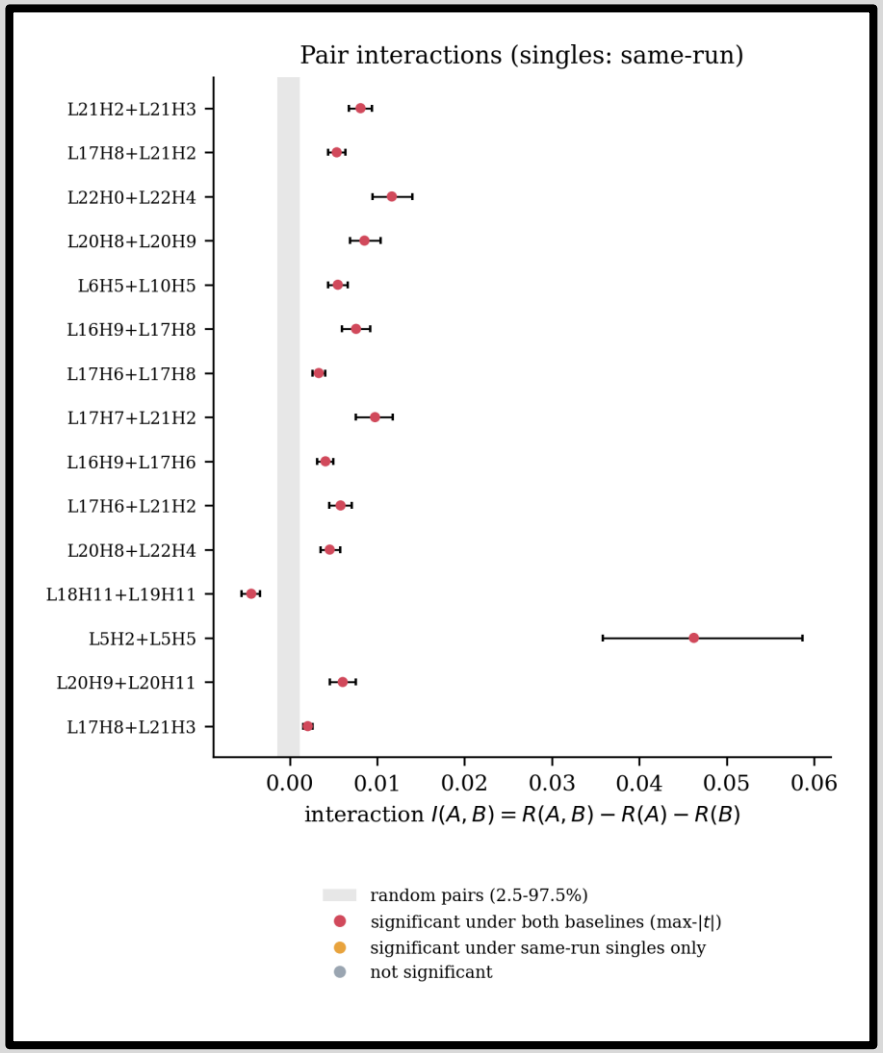


- k=20 heads together: 0.587 vs sum of singles 0.495
- k=100: 0.949 vs 1.053 (saturation, redundancy)
- 100 random heads recover only 0.040
- 137 of 1,225 shortlisted pairs robust; 17 negative
- Necessity and sufficiency correlate only 0.62
- H2 not supported

stage 1: necessity and pairs



necessity vs sufficiency, 336 heads



- L21H0 removes 0.070 of the contrast when corrupted but recovers only 0.015 alone
- Strongest pair: L21H2+L21H3 (I=0.0078, t=11.2)

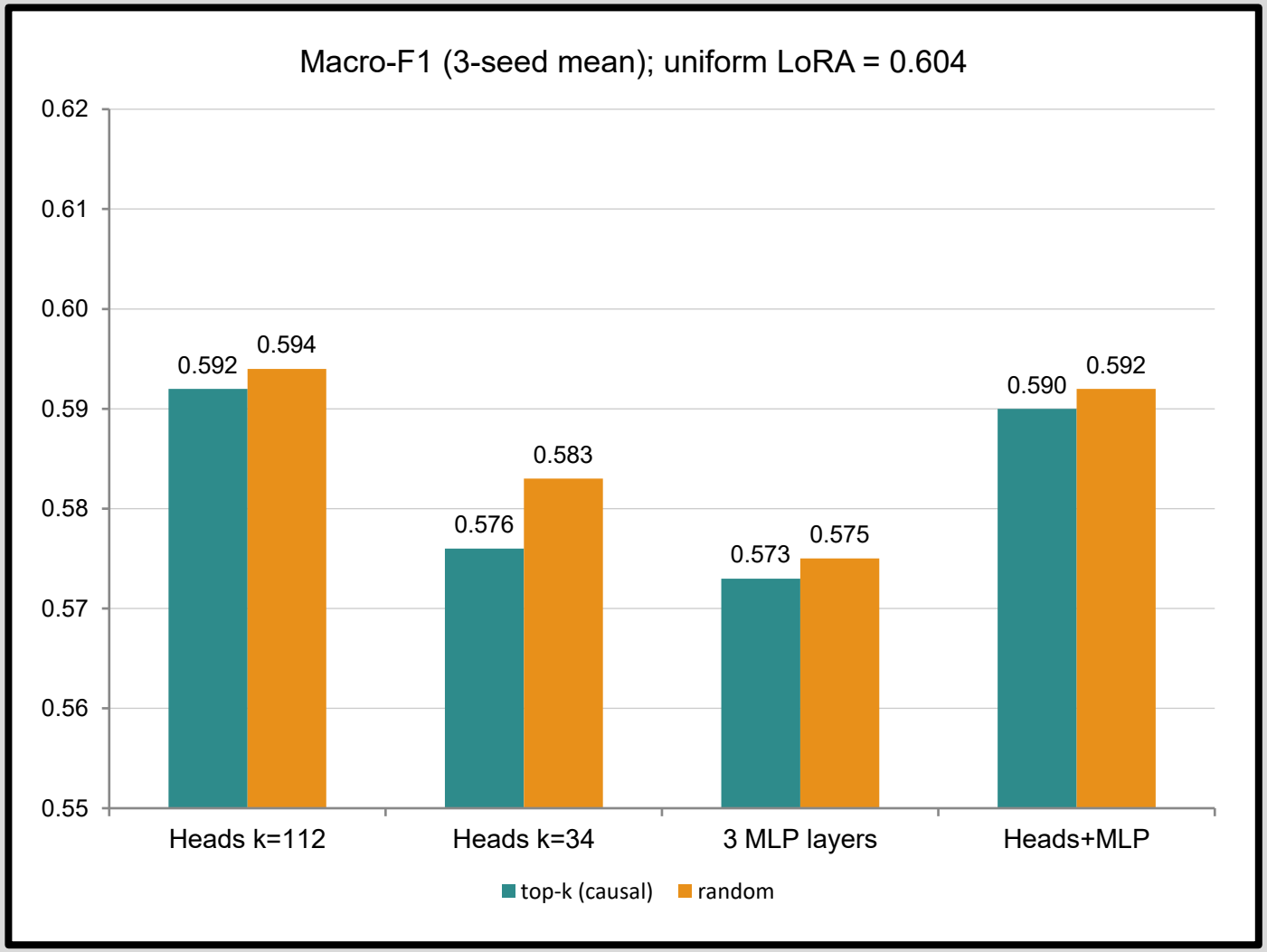
stage 2: circuit routing adapter (CRA)

$$h = W_o x + (\alpha/r) \cdot \Delta(x; G)$$

Fixed, non-trainable mask G scales only the LoRA update; frozen weights are never edited. Applied to q_proj and o_proj (head masks) and MLP down_proj (layer gates).

- Arms: uniform LoRA, top-k by patching score, random with same k (112 or 34 heads), 3 MLP layers, heads + MLP
- Rank 8, α=16, lr 1e-4, 600 steps, batch 32; seeds 0, 1, 2
- Metric: macro-F1 on SAIL-2017 validation; zero-shot base = 0.319

stage 2: results



- Top-k never beats random: differences −0.002 [−0.009, +0.004] (heads), −0.001 (MLP), −0.002 (both)
- At k=34 top-k is below random in every seed; CI [−0.014, +0.000]
- Axis starts at 0.55 to show small gaps

verdict on H3 and H4

H3 (routing beats random): not supported for heads, MLP layers or both.

H4 (efficiency): partial. Top-k (112 heads) 0.592 vs uniform 0.604, mean −0.012 [−0.020, −0.004].

Mask-active parameters fall to 917,504 vs 1,376,256, but all attention adapters allocate the same 1,376,256 parameters, so stored size is unchanged.

Likely reasons (untested): word-retrieval scores may not predict where sentiment fine-tuning helps; heads interact; budget so large that almost any placement works.

Non-significant differences do not prove the methods are equivalent.

limitations and future work

Limitations

- 39 concepts, hand-chosen; sets overlap; one-author Hindi check
- Patching at last token on retrieval; Stage 2 is comprehension
- One model, one task, three seeds; bootstrap ignores training variance
- Top-k uses signed scores; pairs tested only on a 50-head shortlist

Future work

- Neuron-level MLP patching
- Routing by score magnitude
- More seeds, smaller budgets
- Scoring on a task closer to sentiment
- Other languages and models

the end.